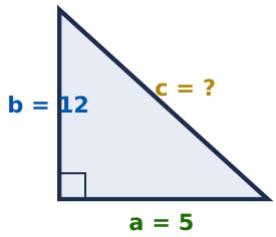
Unit 5 Homework — Pythagorean Theorem — Answer Key

Complete independently · Hypotenuse · Missing Leg · Converse · Word Problems · Coordinate Distance

A. Find the Hypotenuse

Problem 1 Score 3

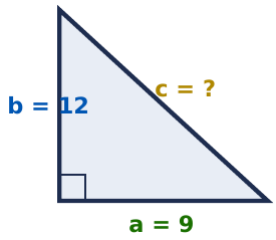
Find the hypotenuse (c) of the right triangle shown.



Answer: 13

Problem 2 Score 3

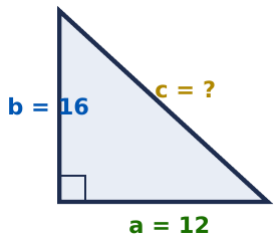
Find the hypotenuse (c) of the right triangle shown.



Answer: 15

Problem 3 Score 4

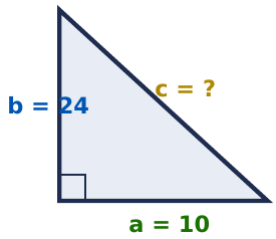
Find the hypotenuse (c) of the right triangle shown.



Answer: 20

Problem 4 Score 4

Find the hypotenuse (c) of the right triangle shown.

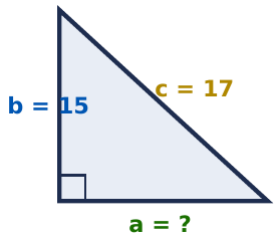


Answer: 26

B. Find a Missing Leg

Problem 1 Score 3

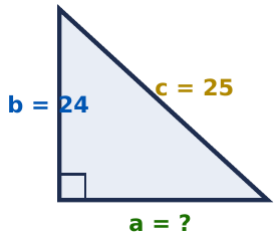
Find the missing leg (a) of the right triangle shown.



Answer: 8

Problem 2 Score 4

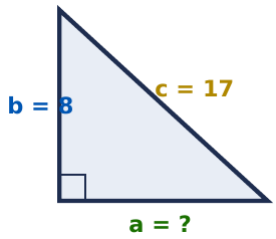
Find the missing leg (a) of the right triangle shown.



Answer: 7

Problem 3 Score 4

Find the missing leg (a) of the right triangle shown.

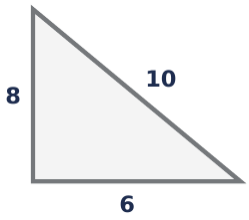


Answer: 15

C. Converse — Is It a Right Triangle?

Problem 1 Score 3

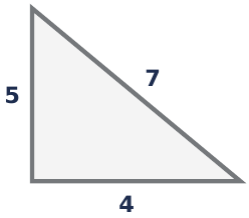
The triangle shown has sides 6, 8, 10. Is it a right triangle?



Answer: yes

Problem 2 Score 4

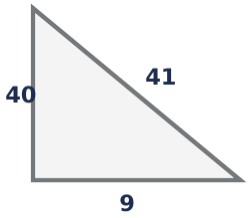
The triangle shown has sides 4, 5, 7. Is it a right triangle?



Answer: no

Problem 3 Score 4

The triangle shown has sides 9, 40, 41. Is it a right triangle?



Answer: yes

Problem 4 Score 5

A triangle has sides 10, 24 and 26. Is it a right triangle? Pick the best reason.

A. Yes. $10^2 + 24^2 = 676$, and $26^2 = 676$.

B. No. 10 + 24 = 34, not 26.

C. Yes. All three sides are even numbers.

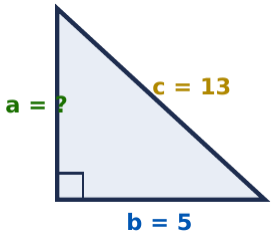
D. No. $10^2 + 26^2$ does not equal 24^2.

Answer: A

D. Word Problems

Problem 1 Score 4

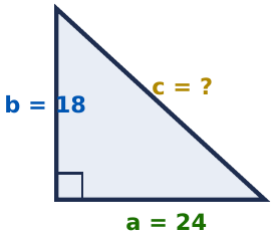
A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high up the wall does the ladder reach?



Answer: 12

Problem 2 Score 5

A TV screen is 24 inches wide and 18 inches tall. What is the diagonal length?



Answer: 30

Problem 3 Score 5

A box is 2 cm wide, 3 cm long and 6 cm tall. Find the length of the longest diagonal inside the box, in cm. (Find the diagonal of the bottom first.)

Answer: 7

Problem 4 Score 5

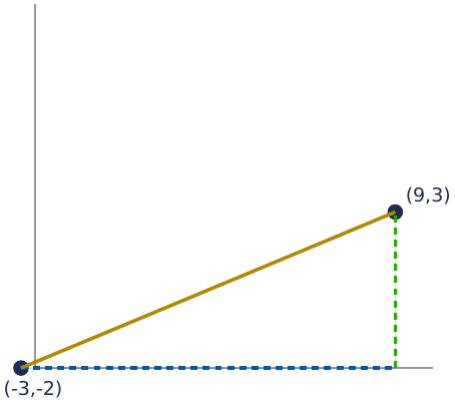
A 25 ft ladder leans on a wall with its base 7 ft from the wall. The base slides out until it is 15 ft from the wall. How many feet does the top of the ladder slide DOWN?

Answer: 4

E. Distance on the Coordinate Plane

Problem 1 Score 4

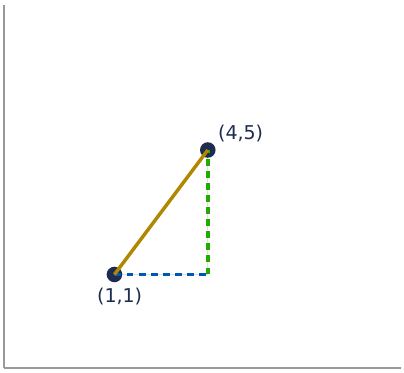
Find the distance between the two points shown.



Answer: 13

Problem 2 Score 5

Find the distance between the two points shown.

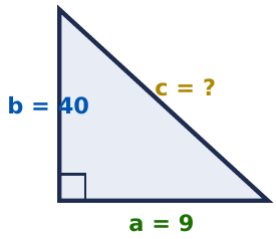


Answer: 5

F. Mixed Review

Problem 1 Score 4

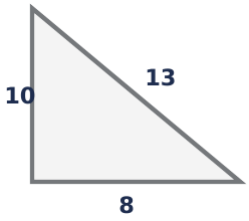
Find the hypotenuse (c) of the right triangle shown.



Answer: 41

Problem 2 Score 4

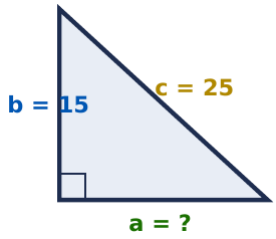
The triangle shown has sides 8, 10, 13. Is it a right triangle?



Answer: no

Problem 3 Score 5

Find the missing leg (a) of the right triangle shown.



Answer: 20